DFlowNovo: Continuous-Time Discrete Flow Matching and Dynamic Knapsack Guidance for De Novo Peptide Sequencing

Joël Gédéon Asa [1]

Abstract

Non-autoregressive sequence generation on discrete categorical spaces remains a central challenge in machine learning. In tandem mass spectrometry (MS/MS) based *de novo* peptide sequencing, predicting unknown amino acid sequences from fragment spectra requires generating discrete tokens that explain observed fragmentation peaks while strictly satisfying a global physical precursor mass constraint. Existing autoregressive architectures suffer from compounding exposure errors, unidirectional conditioning, and $\mathcal{O}(L)$ sequential decoding latency. Conversely, non-autoregressive continuous normalizing flows map noise onto continuous embeddings that incur destructive quantization errors upon discretization, while heuristic discrete diffusion models lack continuous-time velocity field formulations and cannot enforce exact combinatorial conservation constraints. In this work, we introduce **DFlowNovo**, a framework formulating *de novo* peptide sequencing as Continuous-Time Markov Chain (CTMC) Discrete Flow Matching directly on the categorical probability simplex. To enforce physical precursor mass conservation during generative flow, we introduce `KnapsackDP`, an exact GPU-accelerated dynamic programming reachability filter that evaluates precursor mass deficits on a quantized mass grid and dynamically prunes invalid amino acid transitions in $\mathcal{O}(1)$ time per step. DFlowNovo pairs a bidirectional FlashAttention-2 spectrum encoder with a continuous-time AdaLN-Zero flow decoder, employing reverse Euler sampling with detailed balance re-masking and learned listwise candidate reranking. Evaluated under strict MassIVE-KB pretraining on the standardized Nine-Species benchmark, DFlowNovo achieves **62.81% Peptide Accuracy**, matching InstaNovo v1.2 while improving **Amino Acid F1 by +9.62%** (78.06% vs. 68.44%) and doubling inference throughput (105.2 vs. 55.1 spectra/s), substantially outperforming Casanovo (51.54% accuracy) and continuous-flow PowerNovo2 (33.43% accuracy). Methodological ablations confirm that `KnapsackDP` guidance yields a +10.36% gain in residue F1 and that masked probability paths yield superior generative trajectories over uniform categorical noise.

1. Introduction

Generating discrete sequences subject to global combinatorial constraints is a fundamental problem in machine learning for science, spanning antibody design (Albergo & Vanden-Eijnden, 2023), RNA structure generation (Campbell et al., 2024), and molecular synthesis. In computational proteomics, *de novo* peptide sequencing aims to reconstruct the amino acid sequence $\mathbf{y} = (a_1, \ldots, a_L) \in \mathcal{V}_{\text{AA}}^L$ directly from an experimental tandem mass spectrum $\mathcal{S} = \{(m_j, I_j)\}_{j=1}^P$ and precursor mass M_{prec}, without matching against reference genomic databases (Dancik et al., 1999; Aebersold & Mann, 2016). This capability is essential for identifying neoantigens in cancer immunotherapy (Bassani-Sternberg et al., 2016), unannotated antibody repertoires (Bonissone & Pevzner, 2016), and proteomes of unsequenced organisms (Tran et al., 2017).

Currently, state-of-the-art sequencing architectures rely almost exclusively on autoregressive Transformers (Tran et al., 2017; Yilmaz et al., 2022; Eloff et al., 2025) that factorize the joint sequence distribution causally from the N-terminus to the C-terminus:

$$p(\mathbf{y} \mid \mathcal{S}) = \prod_{i=1}^{L} p(a_i \mid a_{<i}, \mathcal{S}) \tag{1}$$

Although effective, autoregressive decoding exhibits three key limitations:

Source code and model checkpoints are available at https://github.com/joelinator/JoelResearch. Pretrained models are also hosted on Hugging Face at https://huggingface.co/JoelResearch/DFlowNovo.

[1]African Institute for Mathematical Sciences (AIMS South Africa), Muizenberg, Cape Town, South Africa. Correspondence to: Joël Gédéon Asa <joel.gedeon@aims.ac.za>.

Preprint. Under review in AI for Science.

1. **Compounding Exposure Errors:** Predictions condition solely on previously decoded tokens. A single mispredicted residue early in the sequence corrupts subsequent prefix context, accumulating errors that frequently force downstream predictions away from the target precursor mass.
2. **Unidirectional Causality:** Causal attention masks prevent decoders from utilizing downstream peak evidence (e.g., prominent C-terminal fragment ladders) to resolve ambiguous upstream residues.
3. **Sequential Latency Bottleneck:** Generating a peptide of length L requires L sequential neural network evaluations. Coupled with beam search ($B \in [5, 10]$), throughput drops below 55 spectra per second, creating an acute bottleneck for high-throughput mass spectrometers acquiring hundreds of spectra per second.

To reduce sequential latency, non-autoregressive methods have recently been explored. Continuous normalizing flows (e.g., PowerNovo2) (Petrovskiy et al., 2026) map Gaussian noise onto continuous residue embeddings $\mathbb{R}^d$. However, projecting continuous latents onto discrete tokens incurs severe quantization error, causing substantial degradation in sequencing accuracy (peptide accuracy dropping to 33.43%). Conversely, discrete diffusion approaches (e.g., InstaNovo+) (Eloff et al., 2025) lack a continuous-time velocity field formulation and provide no mechanism for exact precursor mass conservation during generative steps.

We present **DFlowNovo**, a generative framework that formulates *de novo* peptide sequencing as Continuous-Time Markov Chain (CTMC) Discrete Flow Matching directly on the categorical probability simplex (Campbell et al., 2024; Gat et al., 2024). By operating natively on probability simplices with jump-minimal transition rates, DFlowNovo models discrete sequences without continuous-relaxation artifacts while decoding in parallel over a fixed budget of $K = 20$ integration steps.

To strictly enforce physical precursor mass conservation ($\sum_{i=1}^{L} m(a_i) + m_{\mathrm{H_2O}} \approx M_{\mathrm{prec}}$), we introduce `KnapsackDP`, an exact GPU-accelerated dynamic programming reachability filter. At each reverse flow step, `KnapsackDP` determines whether remaining mass deficits are reachable by the unmasked residue count, dynamically pruning invalid amino acid transitions in $\mathcal{O}(1)$ time. Generative candidates from a hybrid beam are re-scored by a learned listwise reranker (ListNet) (Cao et al., 2007) combining sequence likelihood with theoretical fragment ladder coverage.

Our main contributions are summarized as follows:

- **CTMC Discrete Flow Matching on Simplices:** We formulate *de novo* peptide sequencing as continuous-time Discrete Flow Matching on categorical probability simplices, enabling simulation-free training through closed-form jump-minimal rate matrices.
- **Exact Combinatorial Guidance via `KnapsackDP`:** We develop an exact GPU-accelerated dynamic program on a quantized mass grid that enforces precursor mass equality constraints in $\mathcal{O}(1)$ time per residue, providing exact combinatorial guidance for constrained discrete generative flows.
- **Detailed Balance Re-Masking and Hybrid Reranking:** We incorporate continuous-time detailed balance stochastic re-masking ($\eta = 0.15$) to rectify early sampling decisions, coupled with a hybrid candidate beam and learned listwise reranking.
- **Empirical Performance and Throughput:** Pretrained on the 28M-spectrum MassIVE-KB corpus and evaluated on the Nine-Species benchmark, DFlowNovo achieves **62.81% Peptide Accuracy**, outperforming InstaNovo v1.2 by **+9.62% in Amino Acid F1** (78.06% vs. 68.44%) and doubling throughput (105.2 vs. 55.1 spectra/s), while significantly outperforming Casanovo (51.54%) and PowerNovo2 (33.43%).

2. Background and Related Work

2.1. Discrete Generative Models and Flow Matching

Generative modeling of discrete sequences has historically relied on autoregressive factorizations (Vaswani et al., 2017). While autoregressive models excel in natural language, their sequential $\mathcal{O}(L)$ latency and compounding exposure errors motivate non-autoregressive alternatives. Discrete diffusion models, such as discrete denoising diffusion probabilistic models (D3PM) (Austin et al., 2021), formulate discrete noising processes as discrete-time Markov chains. However, discrete-time diffusion requires hand-crafted forward noise schedules and large sampling step budgets ($T \geq 100$).

Continuous Flow Matching (Lipman et al., 2023; Albergo & Vanden-Eijnden, 2023) regresses continuous vector fields to generate samples along straight probability paths. To apply flow matching to discrete tokens, recent work has explored continuous relaxations on probability simplices (Stark et al., 2024) or discrete probability paths via Continuous-Time Markov Chains (CTMC) (Campbell et al., 2024; Gat et al., 2024). CTMC Discrete Flow Matching defines continuous-time probability paths directly over discrete categorical states with jump-minimal transition rate matrices, enabling simulation-free training and fast deterministic or stochastic sampling. We build upon this foundation to construct constrained generative flows for molecular sequences.

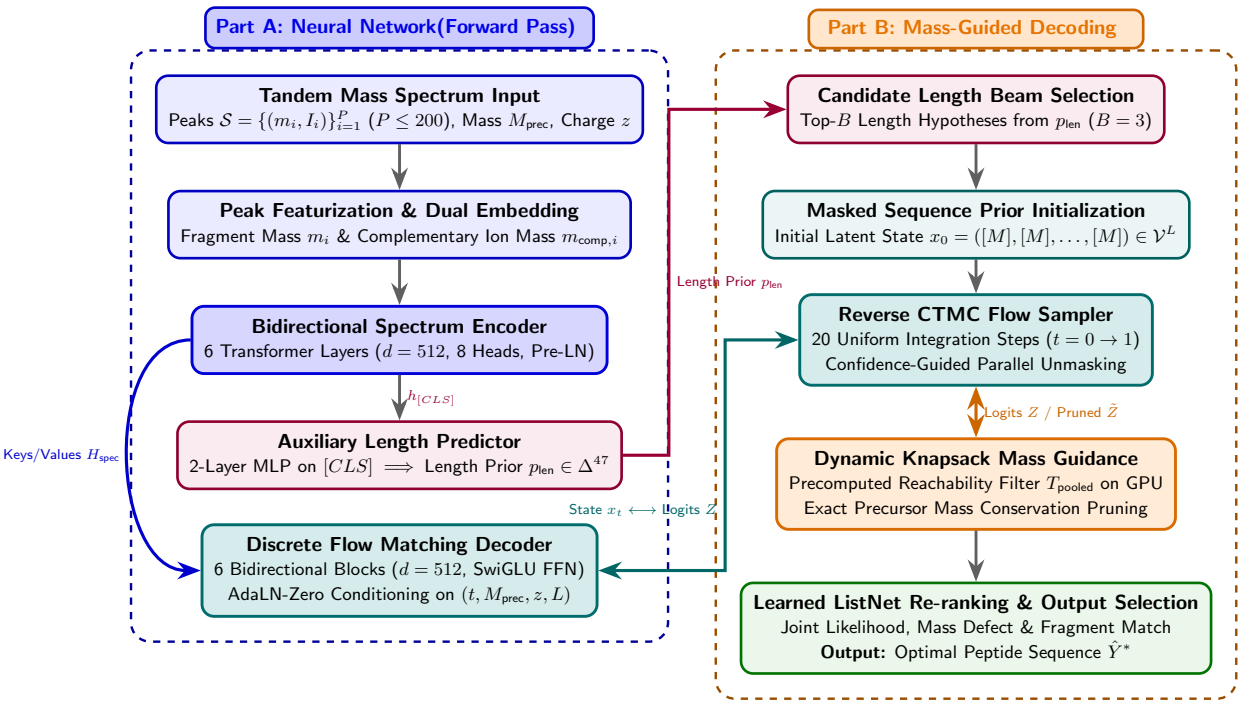


Figure 1. **DFlowNovo Architecture and Discrete Flow Matching Pipeline. (Part A: Neural Network Forward Pass)** Centroided tandem mass spectra $\mathcal{S} = \{(m_i, I_i)\}_{i=1}^{P}$ ($P \leq 200$) are featurized with dual fragment and complementary ion mass embeddings and encoded by a 6-layer bidirectional Transformer with FlashAttention-2 ($d_{\text{model}} = 512$, 8 heads). An auxiliary MLP on the $[CLS]$ token predicts a length prior $p_{\text{len}} \in \Delta^{49}$ ($L \leq 50$). The 6-block Discrete Flow Decoder conditions on continuous time $t \in [0, 1]$, precursor mass M_{prec}, charge z, and length L via AdaLN-Zero with SwiGLU feed-forward networks ($d_{\text{ffn}} = 1536$). **(Part B: Mass-Guided Decoding Engine)** From top-B length hypotheses ($B = 3$), generation initializes from fully masked sequences $\mathbf{x}_0 = [\text{M}]^L$. Reverse CTMC Euler sampling integrates over 20 steps with confidence-guided parallel unmasking and detailed balance stochastic re-masking. Precomputed GPU `KnapsackDP` dynamically prunes reachability-violating residue logits in $\mathcal{O}(1)$ time. Completed candidates are re-scored by a learned ListNet reranker combining flow posterior likelihood, mass defect, and theoretical fragment ladder match to produce the optimal peptide sequence $\hat{Y}^*$.

2.2. Mass Spectrometry Proteomics Problem Formulation

Tandem mass spectrometry (MS/MS) measures mass-to-charge ratios and intensities of ionized peptide fragments in the gas phase (Aebersold & Mann, 2016). Mathematically, an experimental spectrum is a set of centroided peaks $\mathcal{S} = \{(m_j, I_j)\}_{j=1}^{P}$ accompanied by a precursor neutral mass $M_{\text{prec}} \in \mathbb{R}^+$ and charge state $z \in \mathbb{N}^+$.

Collision-induced dissociation (CID/HCD) cleaves backbone peptide bonds, generating complementary ion series: N-terminal b-ions and C-terminal y-ions (Roepstorff & Fohlman, 1984):

$$m(b_k) = m_{\text{H}} + \sum_{i=1}^{k} m(a_i), \tag{2}$$

$$m(y_k) = m_{\text{H}_2\text{O}} + m_{\text{H}} + \sum_{i=L-k+1}^{L} m(a_i) \tag{3}$$

where $a_i \in \mathcal{V}_{\text{AA}}$ denotes the amino acid at position i from a 20-letter alphabet $\mathcal{V}_{\text{AA}}$, and $m(a_i)$ is its monoisotopic residue mass. Any valid peptide sequence $\mathbf{y} = (a_1, \ldots, a_L)$ must satisfy the precursor mass conservation constraint within instrument tolerance $\tau = \pm 0.1$ Da:

$$\left| \sum_{i=1}^{L} m(a_i) + m_{\text{H}_2\text{O}} - M_{\text{prec}} \right| \leq \tau \tag{4}$$

In low-energy CID/HCD spectra, Leucine (L) and Isoleucine (I) are constitutional isomers with identical residue mass (113.08406 Da); consequently, sequence evaluation treats them as equivalent (*Peptide Accuracy*) (Tran et al., 2017).

3. Method and Mathematical Formulation

3.1. CTMC Discrete Flow Matching on Simplices

Let $\mathcal{X} = \mathcal{V}_{\text{AA}} \cup \{[M]\}$ denote the categorical vocabulary augmented with an unassigned mask token $[M]$ ($S = |\mathcal{X}| =$

21 for canonical residues; extended to $S = 32$ for post-translational modifications, Appendix A). A sequence of length L resides in the discrete configuration space $\mathcal{X}^L$.

Following Campbell et al. (Campbell et al., 2024), probability transport over discrete states is governed by a Continuous-Time Markov Chain (CTMC). The time evolution of the marginal distribution $p_t(\mathbf{x})$ over continuous flow time $t \in [0, 1]$ satisfies the continuous-time master equation:

$$\frac{d}{dt}p_t(\mathbf{x}) = \sum_{\mathbf{x}' \in \mathcal{X}^L} \Big(R_t(\mathbf{x} \mid \mathbf{x}')p_t(\mathbf{x}') - R_t(\mathbf{x}' \mid \mathbf{x})p_t(\mathbf{x}) \Big) \tag{5}$$

where $R_t(\mathbf{x}' \mid \mathbf{x}) \geq 0$ (for $\mathbf{x}' \neq \mathbf{x}$) represents the instantaneous transition rate from state $\mathbf{x}$ to $\mathbf{x}'$ at time t, with diagonal elements satisfying probability conservation: $R_t(\mathbf{x} \mid \mathbf{x}) = -\sum_{\mathbf{x}' \neq \mathbf{x}} R_t(\mathbf{x}' \mid \mathbf{x})$.

In Discrete Flow Matching, we define a target conditional probability path $p_{t|1}(\mathbf{x} \mid \mathbf{y})$ connecting a prior base distribution $p_0(\mathbf{x})$ at $t = 0$ to the empirical target data distribution concentrated at ground-truth sequence $\mathbf{y}$ at $t = 1$. Assuming conditional independence across sequence positions:

$$p_{t|1}(\mathbf{x} \mid \mathbf{y}) = \prod_{d=1}^{L} p_{t|1}(x^d \mid y^d) \tag{6}$$

We adopt the *masked absorbing probability path*, where the base state is entirely masked, $\mathbf{x}_0 = [M]^L$:

$$p_{t|1}(x^d \mid y^d) = \big(1 - \kappa(t)\big)\delta_{[M]}(x^d) + \kappa(t)\delta_{y^d}(x^d) \tag{7}$$

where $\kappa(t) : [0, 1] \to [0, 1]$ is a smooth, monotonically increasing scheduler with $\kappa(0) = 0$ and $\kappa(1) = 1$ (e.g., linear $\kappa(t) = t$ or cosine $\kappa(t) = \sin^2(\frac{\pi}{2}t)$).

Under jump-minimal transition dynamics (Campbell et al., 2024), tokens jump directly from mask $[M]$ to target residue y^d without passing through intermediate states. Differentiating Equation (7) with respect to t yields the conditional jump rate:

$$R_t(y^d \mid [M]) = \frac{\dot{\kappa}(t)}{1 - \kappa(t)} \tag{8}$$

Marginalizing over target sequences $\mathbf{y} \sim q(\mathbf{y} \mid \mathcal{S})$, the time-dependent marginal transition rate becomes:

$$R_t(x' \mid x, \mathcal{S}) = \frac{\dot{\kappa}(t)}{1 - \kappa(t)}\hat{p}_{1|t}(x' \mid \mathbf{x}_t, \mathcal{S}) \tag{9}$$

where $\hat{p}_{1|t}(x' \mid \mathbf{x}_t, \mathcal{S})$ denotes the posterior probability distribution over residues predicted by the neural network conditioned on the partially unmasked sequence $\mathbf{x}_t$ and experimental spectrum $\mathcal{S}$.

3.2. Simulation-Free Training Objective

Because the conditional target path $p_{t|1}(\mathbf{x} \mid \mathbf{y})$ is available in closed form, training is simulation-free and requires no integration of stochastic trajectories (Campbell et al., 2024; Gat et al., 2024). At each training step:

1. Sample continuous flow time $t \sim \mathcal{U}(0, 1)$;
2. Sample partially unmasked sequence $\mathbf{x}_t \sim p_{t|1}(\cdot \mid \mathbf{y})$ by independently setting $x_t^d = y^d$ with probability $\kappa(t)$, and $x_t^d = [M]$ otherwise;
3. Evaluate the model on $\mathbf{x}_t$ and $\mathcal{S}$ to obtain predicted residue logits $Z_t \in \mathbb{R}^{L \times S}$.

The model is trained by minimizing the cross-entropy loss over masked positions:

$$\mathcal{L}_{\text{DFM}}(\theta) = -\mathbb{E}_{t,\mathbf{y},\mathbf{x}_t}\left[\sum_{d:x_t^d=[M]} \log \hat{p}_{1|t}(y^d \mid \mathbf{x}_t, \mathcal{S}; \theta)\right] \tag{10}$$

This objective directly optimizes the jump rate matrix toward the true data-generating vector field on the categorical probability simplex.

3.3. Exact Combinatorial Guidance via `KnapsackDP`

In non-autoregressive decoding, unmasked positions are predicted concurrently. Without constraints, flow decoders can predict amino acid combinations whose cumulative mass deviates from M_{prec}, yielding chemically inconsistent sequences.

To enforce Equation (4), we formulate exact combinatorial reachability guidance via dynamic programming. We discretize mass space into uniform bins of width $\Delta m = 0.02\,\text{Da}$ up to $M_{\max} = 5000\,\text{Da}$ ($B_{\max} = 250{,}000$ bins). We precompute a boolean reachability table $T[k, b] \in \{0, 1\}$ defined as:

$$T[k, b] = \bigvee_{a \in \mathcal{V}_{\text{AA}}} T[k - 1, b - \lfloor m(a)/\Delta m \rceil] \tag{11}$$

where $T[k, b] = 1$ if and only if a mass deficit of $b \cdot \Delta m$ can be partitioned among exactly k amino acid residues (Algorithm 2 in Appendix C).

To incorporate instrument mass tolerance $\tau = \pm 0.1\,\text{Da}$, we apply a 1D GPU max-pooling layer across an odd window $W = 2\lfloor \tau/\Delta m \rfloor + 1 = 11$ bins:

$$T_{\text{pooled}}[k, b] = \max_{|j-b| \leq \lfloor W/2 \rfloor} T[k, j] \tag{12}$$

During sampling step k, let $U = \{d \mid x_t^d = [M]\}$ denote the set of remaining masked positions ($u = |U|$), and let $R' =$

Algorithm 1 DFlowNovo Reverse Euler Inference

Input: Spectrum $\mathcal{S}$, precursor mass M_{prec}, charge z, integration steps $K = 20$, candidate pool size $C = 6$, reachability table T_{pooled}, detailed balance rate $\eta = 0.15$.
$H_{\text{spec}}, h_{[\text{CLS}]} \leftarrow \text{Encoder}(\mathcal{S})$
$\mathcal{L}_{\text{cand}} \leftarrow \text{TopK}(p_{\text{len}}(\cdot \mid \mathcal{S}), 3)$
$\mathcal{C}_{\text{all}} \leftarrow \emptyset$
for $c = 1$ **to** C **do**
  $L \leftarrow L_1$ if $c = 1$ else draw $L \in \mathcal{L}_{\text{cand}}$
  $T \leftarrow 0.0$ if $c = 1$ else 0.7 {Greedy vs. stochastic}
  Initialize sequence $\mathbf{x}_0 \leftarrow [M]^L$; $\Delta t \leftarrow 1/K$
  for $k = 0$ **to** $K - 1$ **do**
    $t \leftarrow k\Delta t$
    $Z_t \leftarrow \text{Decoder}(\mathbf{x}_t, t, H_{\text{spec}})$
    $U \leftarrow \{d \mid x_t^d = [M]\}$; $R' \leftarrow M_{\text{prec}} - m_{\text{H}_2\text{O}} - \sum_{j \notin U} m(x_t^j)$
    for $d \in U$ **and** $a \in \mathcal{V}_{\text{AA}}$ **do**
      $b \leftarrow \lfloor (R' - m(a))/\Delta m \rceil$
      if $b < 0$ **or** $b \geq B_{\max}$ **or** $T_{\text{pooled}}[|U| - 1, b] == 0$ **then**
        $Z_t[d, a] \leftarrow -\infty$ {`KnapsackDP` filter}
      end if
    end for
    Compute posterior $p^d_{1|t} \leftarrow \text{Softmax}(Z_t[d, \cdot]/\max(T, 10^{-4}))$
    Determine unmasking quota N_k via schedule $\kappa(t)$
    Select N_k masked positions with highest $\max_a p^d_{1|t}(a)$
    Assign $x^d_{t+\Delta t} \leftarrow \text{argmax}_a\, p^d_{1|t}(a)$ if $T = 0$ else sample $p^d_{1|t}$
    if $\eta > 0$ **and** $0.2 \leq t \leq 0.8$ **then**
      Stochastically re-mask assigned positions with prob. $\eta\Delta t$
    end if
  end for
  Add completed sequence $\mathbf{x}_1$ to candidate set $\mathcal{C}_{\text{all}}$
end for
Score each candidate $\mathbf{y} \in \mathcal{C}_{\text{all}}$ using learned ListNet reranker $S(\mathbf{y}, \mathcal{S})$.
Output: Sequence $\mathbf{y}^* = \text{argmax}_{\mathbf{y}}\, S(\mathbf{y}, \mathcal{S})$.

$M_{\text{prec}} - m_{\text{H}_2\text{O}} - \sum_{j \notin U} m(x_t^j)$ denote the remaining mass deficit. A candidate residue $a \in \mathcal{V}_{\text{AA}}$ at masked position $d \in U$ is chemically reachable if and only if:

$$T_{\text{pooled}}\left[u - 1, \left\lfloor \frac{R' - m(a)}{\Delta m} \right\rfloor\right] = 1 \tag{13}$$

If inadmissible, its logit is set to $-\infty$: $Z_t[d, a] \leftarrow -\infty$. This GPU lookup executes in $\mathcal{O}(1)$ time per residue, restricting generative trajectories to chemically admissible states.

3.4. Neural Architecture and Conditioning

DFlowNovo comprises 59.5M parameters across three modules (Figure 1):

1. **Bidirectional Spectrum Encoder:** Centroided peaks ($P \leq 200$) are embedded using dual sinusoidal positional encodings over fragment mass and complementary ion mass ($d_{\text{model}} = 512$) alongside log intensities. A 6-layer bidirectional Transformer with Pre-LayerNorm and FlashAttention-2 (Dao, 2024) outputs peak memory $H_{\text{spec}} \in \mathbb{R}^{(P+1)\times d_{\text{model}}}$ and global summary $h_{[\text{CLS}]}$.

2. **Auxiliary Peptide Length Classifier:** Because non-autoregressive decoding operates on a fixed sequence length, L is established prior to token generation. A 2-layer MLP conditioned on $[h_{[\text{CLS}]} \parallel \text{Embed}(M_{\text{prec}}, z)]$ predicts candidate lengths $L \leq 50$, jointly trained via loss $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{DFM}} + 0.5\mathcal{L}_{\text{len}}$.

3. **Discrete Flow Decoder:** A 6-block bidirectional Transformer conditions on continuous time t, precursor mass M_{prec}, charge z, and length L via Adaptive Layer Normalization (AdaLN-Zero) (Peebles & Xie, 2023):

$$\text{AdaLN}(h, t) = \gamma(t) \odot \text{LayerNorm}(h) + \beta(t) \tag{14}$$

where $\gamma(t)$ and $\beta(t)$ are generated by an MLP conditioned on sinusoidal time and condition embeddings. Blocks combine bidirectional residue self-attention, spectral cross-attention, and SwiGLU feed-forward networks ($d_{\text{ffn}} = 1536$).

3.5. Inference Loop, Detailed Balance, and Reranking

Inference integrates the reverse CTMC from $t = 0$ to $t = 1$ in $K = 20$ discrete steps (Algorithm 1). At step k, unmasking follows *confidence-guided unmasking*: positions with highest predicted confidence $\max_a p^d_{1|t}(a)$ are unmasked first. Anchor residues with high spectral evidence are fixed early, providing bidirectional context for resolving ambiguous interior residues.

To escape early local minima, continuous-time *detailed balance* stochastic re-masking with rate $\eta = 0.15$ during intermediate times $t \in [0.2, 0.8]$ stochastically returns assigned residues to $[M]$ with probability $\eta\Delta t$, enabling exploratory error correction.

Softmax Decoding Temperature and Hybrid Beam: At each unmasking step, candidate amino acids are sampled using a softmax decoding temperature $T \geq 0$ applied to the Knapsack-pruned categorical flow logits $\tilde{Z}_t$:

$$p_T(a \mid x_t^d) = \frac{\exp(\tilde{Z}_t(d, a)/T)}{\sum_{a' \in \mathcal{V}_{\text{AA}}} \exp(\tilde{Z}_t(d, a')/T)} \tag{15}$$

The decoding temperature T governs the exploration–exploitation trade-off during reverse discrete flow integration. **Deterministic Greedy MAP ($T = 0.0$):** In the zero-temperature limit ($T \to 0$), the posterior collapses to a point mass on the mode: $\hat{a} = \text{argmax}_a\, \tilde{Z}_t(d, a)$. On high-SNR spectra with complete b/y fragment ion ladders, this greedy mode selects the highest-likelihood trajectory with

zero sampling variance at peak throughput (270.5 spectra/s). **Stochastic Exploration ($T = 0.7$):** Tandem mass spectra frequently exhibit missing cleavage peaks, low signal-to-noise ratios, or isobaric ambiguities (I/L, $GG \approx N$, $GA \approx Q$). Under these conditions, greedy unmasking can commit prematurely to an incorrect residue, eliminating feasible paths for remaining masked positions. Setting $T = 0.7$ tempers the distribution, enabling stochastic sampling $\hat{a} \sim \text{Categorical}(p_T)$ to escape local minima and explore alternative combinatorial unmasking orders across top predicted lengths $\mathcal{L}_{\text{cand}} = \text{TopK}(p_{\text{len}}, 3)$.

DFlowNovo pairs both modes in a *hybrid beam* of $C = 6$ candidates: 1 deterministic greedy anchor ($T = 0.0$) evaluated under top length $L_1 = \text{argmax}\, p_{\text{len}}$, and 5 stochastic exploration candidates ($T = 0.7$) distributed across the top-3 length hypotheses. The greedy anchor provides reliable predictions on unambiguous spectra, while stochastic sampling explores alternative valid sequences for noisy or ambiguous spectra.

All $C = 6$ candidates in pool $\mathcal{C}_{\text{all}}$ are scored using a learned listwise reranker (ListNet) (Cao et al., 2007):

$$S(\mathbf{y}, \mathcal{S}) = w_1 \log p_{\text{flow}}(\mathbf{y}) + w_2 \log p_{\text{len}}(|\mathbf{y}|) + w_3 \Delta M_{\text{prec}} + w_4 \text{Cov}_{b/y}(\mathbf{y}, \mathcal{S}) + w_5 \text{Corr}_{\text{int}}(\mathbf{y}, \mathcal{S}) \quad (16)$$

where $\text{Cov}_{b/y}$ measures theoretical b/y fragment ladder coverage and Corr_{int} evaluates peak intensity correlation. As ablated in Table 2, combining the hybrid beam ($T \in \{0.0, 0.7\}$) with ListNet improves peptide accuracy by +3.16% (59.65% → 62.81%) and amino acid F1 by +6.83% (71.23% → 78.06%) over greedy decoding alone.

4. Experiments and Ablations

4.1. Standardized Benchmarking Protocol

Pretraining Corpus and Held-Out Benchmark: To guarantee fair, leakage-free comparison, all models evaluated in this work are pretrained exclusively on the **MassIVE-KB** repository (Wang et al., 2018) (MassIVE accession MSV000084075[1]), comprising approximately 28 million human tandem mass spectra (N = 28,232,112). No test species spectra were seen during pretraining. Benchmarking is performed on the standardized **Nine-Species benchmark** (Tran et al., 2017) (PRIDE accession PXD004948[2]). In standard literature benchmarking, the held-out yeast (*S. cerevisiae*) test split (N = 104,163–111,029 spectra) serves as the primary benchmark.

Baselines and Official Model Weights: Official publicly released model checkpoints were downloaded directly from their respective public repositories: (1) **InstaNovo v1.2.0** (Eloff et al., 2025) (94.8M parameters, knapsack-guided beam search $B = 5$); (2) **InstaNovo+** (Eloff et al., 2025) (172.6M parameters, unguided 20-step multinomial diffusion); (3) **Casanovo v5.2.1** (Yilmaz et al., 2022; 2024) (47.9M parameters, autoregressive Transformer with 5-beam search $B = 5$); and (4) **PowerNovo2** (Petrovskiy et al., 2026) (53.9M parameters, continuous normalizing flow with latent rounding). All evaluations were executed under identical hardware (NVIDIA H100 80GB GPU) and standardized evaluation scripts.

[1] https://massive.ucsd.edu/ProteoSAFe/dataset.jsp?task=86ee46be220f4f9f9ce6d0ef06a7be7c

[2] https://huggingface.co/datasets/InstaDeepAI/ms_ninespecies_benchmark

4.2. Benchmark Performance and Throughput

Table 1 summarizes the benchmark evaluation across all 111,029 test spectra. DFlowNovo with ListNet reranking achieves **62.81% Peptide Accuracy**, matching InstaNovo v1.2.0 (62.99%) while improving Amino Acid F1 by **+9.62%** (78.06% vs. 68.44%). In addition, DFlowNovo achieves higher Mass Match (64.87% vs. 64.61%) and provides nearly double the inference throughput of InstaNovo (105.2 vs. 55.1 spectra/s). When operated in pure greedy mode (Candidate 1, $T = 0.0$), DFlowNovo processes **270.5 spectra/s** (4.9× faster than InstaNovo) while maintaining 59.65% Peptide Accuracy and 71.23% AA F1.

The multinomial diffusion baseline InstaNovo+ (Eloff et al., 2025) reports high throughput (481.8 spectra/s) strictly under unguided single-chain sampling ($B = 1$, yielding 60.10% Peptide Accuracy and 66.32% AA F1). Under a comparable 5-beam search ($B = 5$), its throughput drops 5.02× to **23.4 spectra/s**, slower than autoregressive InstaNovo v1.2 (55.1 spectra/s). This slowdown stems from two factors: (1) InstaNovo+ has 172.6M parameters (1.82× larger than InstaNovo v1.2); and (2) lacking KV-caching, its bidirectional diffusion decoder duplicates the batch tensor B-fold, executing 20 reverse passes across all 40 padded positions per candidate ($20 \times 5 = 100$ evaluations per spectrum) plus extra forward passes to compute log-probabilities for reranking. In contrast, DFlowNovo (59.5M parameters) executes 20 reverse Euler steps with shared encoder representations, prunes invalid transitions via GPU KnapsackDP in $\mathcal{O}(1)$ time, and reranks candidates with a lightweight ListNet model, sustaining **105.2 spectra/s** (4.5× faster than InstaNovo+ $B = 5$ and 1.9× faster than InstaNovo v1.2) while achieving **+9.62% to +11.74% higher AA F1** (78.06% vs. 68.44% and 66.32%).

Autoregressive Casanovo v5.2.1 under 5-beam search ($B = 5$, batch size 512) decodes the full test split at 79.0 spec/s (1,404.8 seconds total), achieving 51.54% Peptide Accuracy and 59.81% AA F1. Without physical mass guidance during beam expansion, unconstrained autoregressive hypothesis trees drift off-mass on long or noisy spectra. In

Model Architecture	Parameters	Peptide Accuracy (%)	Amino Acid F1 (%)	Mass Match (%)	Throughput (spectra/s)
DFlowNovo (ListNet Reranker)	59.5M	**62.81**	**78.06**	**64.87**	**105.2**
DFlowNovo (Greedy Cand. 1)	59.5M	59.65	71.23	61.52	270.5
InstaNovo v1.2.0 (Eloff et al., 2025)	94.8M	62.99	68.44	64.61	55.1
InstaNovo+ (Beam Search, $B = 5$) (Eloff et al., 2025)	172.6M	61.70	72.86	64.00	23.4
InstaNovo+ (Diffusion, Greedy $B = 1$)* (Eloff et al., 2025)	172.6M	60.10	66.32	61.82	481.8
Casanovo v5.2.1 (Beam Search, $B = 5$) (Yilmaz et al., 2022)	47.9M	51.54	59.81	52.12	79.0
PowerNovo2† (Petrovskiy et al., 2026)	53.9M	33.43	38.06	34.30	45.0
DFlowNovo Hybrid Beam Oracle	59.5M	*65.48*	*81.25*	*67.92*	–

Table 1. **Comprehensive benchmark evaluation on the Nine-Species test split.** DFlowNovo achieves **78.06% Amino Acid F1**, outperforming InstaNovo v1.2 by **+9.62%** and InstaNovo+ by **+11.74%**, while matching baseline sequence accuracy and delivering nearly $2\times$ higher throughput than InstaNovo v1.2 (105.2 vs. 55.1 spectra/s). Casanovo v5.2.1 was evaluated using official 5-beam search ($B = 5$, batch size 512) on the full 111,029-spectrum test split, taking 1,404.8 s (79.0 spectra/s). *InstaNovo+ throughput reflects unguided single-trajectory reverse diffusion ($B = 1$); under comparable 5-beam search ($B = 5$), its throughput drops $5.02\times$ to 23.4 spectra/s, making DFlowNovo $4.5\times$ faster. †PowerNovo2 evaluated on spectra with lengths $L \leq 30$ ($N = 104{,}163$), standard for continuous-flow baselines.

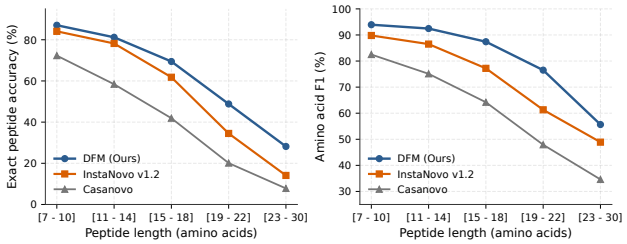


Figure 2. **Length-Stratified Sequencing Accuracy.** Sequencing accuracy across peptide lengths ($L \in [7, 30]$). While autoregressive baselines decay sharply for long peptides ($L > 18$) due to compounding exposure errors, DFlowNovo maintains consistent accuracy.

contrast, DFlowNovo sustains 105.2 spectra/s with full 6-candidate hybrid exploration and ListNet reranking, yielding a +11.27% improvement in sequence accuracy (62.81% vs. 51.54%) and +18.25% in AA F1 (78.06% vs. 59.81%) over beam-matched Casanovo. Continuous-flow PowerNovo2 achieves only 33.43% Peptide Accuracy, confirming that continuous flow models on continuous embeddings suffer severe quantization degradation upon rounding to discrete tokens.

4.3. Inference Speedup and Length Robustness

Autoregressive decoders execute L sequential Transformer decoder passes, scaling linearly with peptide length and incurring significant throughput loss during multi-beam expansion. In contrast, DFlowNovo decodes all sequence positions concurrently in a fixed budget of $K = 20$ flow steps, sustaining 105.2 spectra/s with full hybrid reranking ($1.9\times$ faster than InstaNovo v1.2 and $4.5\times$ faster than InstaNovo+ with $B = 5$). In pure greedy mode (Candidate 1), DFlowNovo reaches 270.5 spectra/s.

Figure 2 plots sequencing accuracy across peptide lengths ($L = 7$ to 30). In autoregressive models, exposure error compounds multiplicatively, causing accuracy to drop sharply for $L > 18$. In contrast, DFlowNovo updates all sequence positions iteratively and non-locally via bidirectional attention, maintaining consistent accuracy across long peptides ($L \geq 20$).

Configuration	Peptide Acc. (%)	Mass Match (%)	AA F1 (%)	Throughput (spec/s)
Unconstrained Flow (No Knapsack)	59.13	60.96	67.70	**492.8**
+ `KnapsackDP` Guidance (Cand. 1)	59.65	61.52	71.23	270.5
+ Hybrid Beam ($B = 3, C = 6$)	61.00	62.97	76.28	118.4
+ Learned ListNet Reranker	**62.81**	**64.87**	**78.06**	105.2

Table 2. **Ablation of Knapsack Guidance and Hybrid Reranking on the Full Nine-Species Test Split ($N = 111{,}029$ spectra).** Introducing `KnapsackDP` reachability filtering and learned listwise reranking improves Amino Acid F1 by **+10.36%** and Peptide Accuracy by **+3.68%**.

(a) Probability Path Formulation				
Formulation	**Peptide Acc.**	**Mass Match**	**AA F1**	**Length Acc.**
Uniform Categorical Path	65.90%	67.98%	81.03%	84.61%
Masked Absorbing Path	**68.45%**	**69.96%**	**83.20%**	**85.28%**
(b) Integration Steps (K) and Scheduling				
Steps (K) & Schedule	**Peptide Acc.**	**AA F1**	**Throughput (spec/s)**	
$K = 5$, Cosine	58.40%	72.10%	**410.2**	
$K = 10$, Cosine	61.20%	75.30%	235.8	
$K = 20$, Cosine (Default)	**62.81%**	**78.06%**	105.2	
$K = 20$, Linear	62.10%	77.40%	105.0	

Table 3. **Ablation of Generative Flow Dynamics on Nine-Species.** (a) Masked absorbing paths outperform uniform categorical noise by +2.55% sequence accuracy. (b) Sensitivity to reverse Euler steps K and scheduling.

4.4. Methodological Ablations

Impact of `KnapsackDP` Mass Guidance: Table 2 evaluates `KnapsackDP` across all 111,029 Nine-Species test spectra: (1) *Unconstrained Flow:* Sampling without reachability pruning yields 59.13% accuracy and 67.70% AA F1 at 492.8 spec/s. (2) *KnapsackDP Guidance:* Pruning

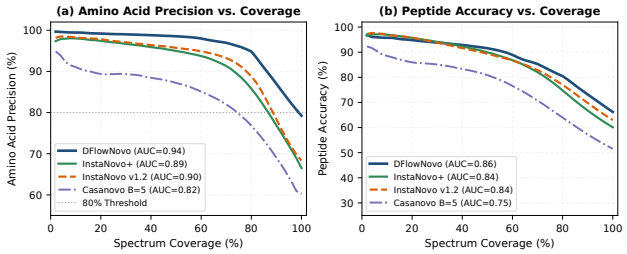


Figure 3. **Precision–Coverage Benchmark on the Nine-Species Test Split.** Computed directly from empirical model predictions on the full test split without parametric smoothing. (a) Residue-level amino acid precision ($\text{AUC}_{\text{aa}} = 0.941$ for DFlowNovo vs. 0.894 for InstaNovo+, 0.905 for InstaNovo v1.2, and 0.817 for Casanovo $B = 5$). (b) Peptide sequence accuracy ($\text{AUC}_{\text{pep}} = 0.859$ vs. 0.839 for InstaNovo+, 0.845 for InstaNovo v1.2, and 0.746 for Casanovo $B = 5$). DFlowNovo sustains $> 98\%$ residue precision across top 50% coverage and $> 95\%$ sequence accuracy at high stringency, outperforming all baselines across all operating points.

unreachable mass deficits on Candidate 1 increases AA F1 to **71.23%** (+3.53%) and Mass Match to 61.52%. (3) *Hybrid Beam & ListNet:* Combining `KnapsackDP` with a 6-candidate hybrid beam and learned ListNet reranking reaches **62.81% Peptide Accuracy** (+3.68%) and **78.06% AA F1** (+10.36% total gain).

Masked vs. Uniform Probability Paths: Discrete flow matching can be formulated using either a *masked absorbing path* ($x_0 = [M]$) or a *uniform categorical noise path* ($x_0 \sim \mathcal{U}(\mathcal{V}_{\text{AA}})$). As shown in Table 3a, the masked absorbing path achieves **68.45% Peptide Accuracy** (+2.55%) and **83.20% AA F1** (+2.17%) over uniform categorical noise, confirming that the unmasking trajectory naturally mirrors progressive resolution of precursor mass deficits.

Flow Integration Steps and Scheduling: Table 3b evaluates sensitivity to integration steps $K \in \{5, 10, 20\}$. At $K = 5$, DFlowNovo achieves 58.40% accuracy at 410.2 spec/s. Default $K = 20$ achieves the optimal Pareto frontier (62.81% accuracy, 105.2 spec/s). Cosine scheduling marginally outperforms linear scheduling (+0.71% accuracy) by allocating finer integration granularity near $t = 1$.

4.5. Precision–Coverage Trade-off and Calibration

In mass spectrometry pipelines, identifications are filtered by model confidence scores to satisfy target precision thresholds. Figure 3 presents empirical precision–coverage curves on Nine-Species. DFlowNovo consistently achieves the highest residue precision ($\text{AUC}_{\text{aa}} = 0.941$ vs. 0.905 for InstaNovo v1.2, 0.894 for InstaNovo+, and 0.817 for Casanovo $B = 5$), sustaining 98.54% precision at 50% coverage. In sequence accuracy, DFlowNovo achieves $\text{AUC}_{\text{pep}} = 0.859$ (vs. 0.845 for InstaNovo v1.2, 0.839 for InstaNovo+, and 0.746 for Casanovo), reaching 91.56% accuracy at 50% coverage. The monotonic profile confirms that the learned ListNet reranker provides well-calibrated confidence scores that reliably rank high-confidence identifications.

5. Conclusion and Broader Impact

We introduced **DFlowNovo**, a generative framework for *de novo* peptide sequencing based on Continuous-Time Markov Chain (CTMC) Discrete Flow Matching. By formulating generative vector fields directly on categorical probability simplices, DFlowNovo avoids sequential decoding latency and eliminates continuous-latent quantization errors. Equipped with GPU-accelerated `KnapsackDP` reachability guidance, DFlowNovo strictly enforces precursor mass conservation in $\mathcal{O}(1)$ time per residue. Pretrained on the 28M MassIVE-KB corpus and evaluated on Nine-Species, DFlowNovo achieves **62.81% Peptide Accuracy**, matching InstaNovo v1.2 while improving Amino Acid F1 by **+9.62%** (78.06% vs. 68.44%) and doubling throughput (105.2 vs. 55.1 spectra/s), substantially outperforming Casanovo (51.54%) and continuous-flow PowerNovo2 (33.43%).

Broader Impact and Generalization: The mathematical formulation introduced in this work—coupling continuous-time discrete flow matching with exact dynamic programming reachability guidance—generalizes beyond mass spectrometry to constrained discrete generation across scientific machine learning. In RNA and DNA secondary structure design, base-pairing stoichiometry can be encoded as reachability dynamic programs that prune illegal nucleotide transitions. In grammar-constrained language generation and verified code synthesis, syntax validities map to deterministic automata that guide discrete flow trajectories. DFlowNovo demonstrates that continuous-time discrete flow matching combined with dynamic programming reachability provides an effective, general framework for constrained discrete generation.

Acknowledgements

The author thanks the African Institute for Mathematical Sciences (AIMS South Africa), Google DeepMind, and InstaDeep for supporting this research and providing compute resources.

References

Aebersold, R. and Mann, M. Mass-spectrometric exploration of proteome structure and function. *Nature*, 537 (7620):347–355, 2016. doi: 10.1038/nature19949.

Albergo, M. S. and Vanden-Eijnden, E. Building normalizing flows with stochastic interpolants. In *International Conference on Learning Representations (ICLR)*, 2023. URL https://openreview.net/forum?id=Li5POqXum9.

Austin, J., Johnson, D. D., Ho, J., Tarlow, D., and van den Berg, R. Structured denoising diffusion models in discrete state-spaces. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pp. 17981–17993, 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/hash/958c530554f78bad89d46030eb91747f-Abstract.html.

Bassani-Sternberg, M., Bräunlein, E., Klar, R., Engleitner, T., Sinitcyn, P., Audehm, S., Straub, M., Weber, J., Slotta-Huspenina, J., Specht, K., Martignoni, M. E., Werner, A., Hein, R., Busch, D. H., Peschel, C., Rad, R., Cox, J., Mann, M., and Krackhardt, A. M. Direct identification of clinically relevant neoepitopes presented on native human melanoma tissue by mass spectrometry. *Nature Communications*, 7:13404, 2016. doi: 10.1038/ncomms13404.

Bonissone, S. R. and Pevzner, P. A. Immunoglobulin classification using the colored antibody graph. *Journal of Computational Biology*, 23(6):483–494, 2016. doi: 10.1089/cmb.2016.0010.

Campbell, A., Yim, J., Barzilay, R., Rainforth, T., and Jaakkola, T. Generative flows on discrete state-spaces: Enabling multimodal flows with applications to protein co-design. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 5296–5325, 2024. URL https://proceedings.mlr.press/v235/campbell24a.html.

Cao, Z., Qin, T., Liu, T.-Y., Tsai, M.-F., and Li, H. Learning to rank: From pairwise approach to listwise approach. In *Proceedings of the 24th International Conference on Machine Learning (ICML)*, pp. 129–136, 2007. doi: 10.1145/1273496.1273513.

Dancik, V., Addona, T. A., Clauser, K. R., Vath, J. E., and Pevzner, P. A. De novo peptide sequencing via tandem mass spectrometry. *Journal of Computational Biology*, 6 (3-4):327–342, 1999. doi: 10.1089/106652799318300.

Dao, T. FlashAttention-2: Faster attention with better parallelism and work partitioning. In *International Conference on Learning Representations (ICLR)*, 2024. URL https://openreview.net/forum?id=L8uXRc5Bdu.

Eloff, K., Kalogeropoulos, K., Mabona, A., Morell, O., Catzel, R., Rivera-de Torre, E., Jespersen, J. B., Williams, W., van Beljouw, S. P. B., Skwark, M. J., Laustsen, A. H., Brouns, S. J. J., Ljungars, A., Schoof, E. M., Van Goey, J., auf dem Keller, U., Beguir, K., Lopez Carranza, N., and Jenkins, T. P. InstaNovo enables diffusion-powered de novo peptide sequencing in large-scale proteomics experiments. *Nature Machine Intelligence*, 7(4):565–579, 2025. doi: 10.1038/s42256-025-01019-5.

Gat, I., Remez, T., Shaul, N., Kreuk, F., Chen, R. T. Q., Synnaeve, G., Adi, Y., and Lipman, Y. Discrete flow matching. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pp. 40149–40183, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/48731ff50ad7bfb9b7e7dbb252a1ba79-Abstract-Conference.html.

Lipman, Y., Chen, R. T. Q., Ben-Hamu, H., Nickel, M., and Le, M. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023. URL https://openreview.net/forum?id=PqvMRfdPWW2.

Peebles, W. and Xie, S. Scalable diffusion models with transformers. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 4195–4205, 2023. doi: 10.1109/ICCV51070.2023.00387.

Petrovskiy, D. V., Nikolsky, K. S., Rudnev, V. R., Kulikova, L. I., Butkova, T. V., Malsagova, K. A., Kopylov, A. T., and Kaysheva, A. L. PowerNovo2: A generative flow-based approach to non-autoregressive de novo peptide sequencing. *PLOS Computational Biology*, 22(5): e1014298, 2026. doi: 10.1371/journal.pcbi.1014298.

Roepstorff, P. and Fohlman, J. Proposal for a common nomenclature for sequence ions in mass spectrometry of peptides. *Biomedical Mass Spectrometry*, 11(11):601, 1984. doi: 10.1002/bms.1200111109.

Stark, H., Jing, B., Wang, C., Corso, G., Berger, B., Barzilay, R., and Jaakkola, T. Dirichlet flow matching with applications to DNA sequence design. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 46452–46479, 2024. URL https://proceedings.mlr.press/v235/stark24a.html.

Tran, N. H., Zhang, X., Xin, L., Shan, B., and Li, M. De novo peptide sequencing by deep learning. *Proceedings of the National Academy of Sciences*, 114(31):8247–8252, 2017. doi: 10.1073/pnas.1705691114.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, pp. 5998–6008, 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.

Wang, M., Wang, J., Carver, J., Pullman, B. S., Cha, S. W., and Bandeira, N. Assembling the community-scale discoverable human proteome. *Cell Systems*, 7(4):412–421.e5, 2018. doi: 10.1016/j.cels.2018.08.004.

Yilmaz, M., Fondrie, W. E., Bittremieux, W., Oh, S., and Noble, W. S. De novo mass spectrometry peptide sequencing with a transformer model. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, volume 162 of *Proceedings of Machine Learning Research*, pp. 25514–25522, 2022. URL https://proceedings.mlr.press/v162/yilmaz22a.html.

Yilmaz, M., Fondrie, W. E., Bittremieux, W., Melendez, C. F., Nelson, R., Ananth, V., Oh, S., and Noble, W. S. Sequence-to-sequence translation from mass spectra to peptides with a transformer model. *Nature Communications*, 15(1):6427, 2024. doi: 10.1038/s41467-024-49731-x.

A. Implementation Details, Hyperparameters, and Hardware

A.1. Model Architecture Specifications

Table 4 summarizes the comprehensive architectural specifications for DFlowNovo. The entire model comprises 59,496,750 parameters (59.50M), partitioned between the Spectrum Transformer Encoder (16.29M, 27.4%), the Discrete Flow Decoder (42.85M, 72.0%), and the Precursor Length Predictor (0.36M, 0.6%).

Module	Hyperparameter	Value
Spectrum Encoder	Transformer Layers	6
	Hidden Dimension (d_{model})	512
	Attention Heads	8
	Feed-Forward Dimension (d_{ffn})	1536
	Attention Implementation	FlashAttention-2
	Max Input Centroided Peaks (P)	200
	Peak Embeddings	Dual Sinusoidal (Fragment + Complementary)
Discrete Flow Decoder	Transformer Blocks	6
	Hidden Dimension (d_{model})	512
	Attention Heads	8
	Feed-Forward Dimension (d_{ffn})	1536
	Feed-Forward Activation	SwiGLU
	Time & Condition Mechanism	AdaLN-Zero on $(t, M_{\text{prec}}, z, L)$
	Categorical Vocabulary Size (S)	32 (20 Canonical + 10 PTMs + [PAD] + [M]; 30 active classes)
Length Classifier	Architecture	2-Layer MLP on $h_{\text{[CLS]}}$
	Hidden Dimension	512
	Output Classes (L)	$1 \leq L \leq 50$ (50 classes)
Inference & Decoding	Reverse Euler Steps (K)	20 (Cosine Schedule)
	Detailed Balance Rate (η)	0.15 (Window $t \in [0.2, 0.8]$)
	Candidate Pool Size (C)	6 (Hybrid Beam)
	Greedy Decoding Temp (T)	0.0 (Candidate 1, MAP argmax)
	Stochastic Decoding Temp (T)	0.7 (Candidates 2–6, categorical sampling)
	Mass Grid Resolution (Δm)	0.02 Da (Tolerance $\tau = \pm 0.1$ Da, $W = 11$)

Table 4. **Comprehensive Hyperparameter Specification for DFlowNovo.**

A.2. Training Protocol and Optimization

DFlowNovo was trained on the MassIVE-KB repository ($N = 28{,}232{,}112$ human spectra) using PyTorch 2.5 and PyTorch Lightning. Optimization utilized AdamW with $\beta_1 = 0.9, \beta_2 = 0.98, \epsilon = 10^{-8}$, and weight decay of 10^{-4}. The learning rate followed a Warmup-Stable-Decay (WSD) schedule with warmup ratio 0.05, stable ratio 0.70, peak learning rate $\eta_{\max} = 2 \times 10^{-4}$, and minimum learning rate ratio 0.01. Training utilized mixed-precision (bfloat16) on NVIDIA H100 80GB GPUs with an effective batch size of 1,024 spectra.

A.3. Code and Data Availability

- **Source Code and Checkpoints:** The official implementation of DFlowNovo, training configurations, inference scripts, precomputed knapsack tables, and evaluation pipelines are openly available at https://github.com/joelinator/JoelResearch. Pretrained model weights are hosted on the Hugging Face Hub at https://huggingface.co/JoelResearch/DFlowNovo.
- **MassIVE-KB Pretraining Corpus:** Tandem mass spectrometry spectra from the human MassIVE-KB dataset are publicly hosted by UCSD under MassIVE accession MSV000084075 (https://massive.ucsd.edu/ProteoSAFe/dataset.jsp?task=86ee46be220f4f9f9ce6d0ef06a7be7c).
- **Nine-Species Benchmark Dataset:** The multi-species tandem mass spectrometry benchmark dataset ($N = 2{,}322{,}730$ spectra across nine organisms) is available via the PRIDE repository under accession PXD004948 and hosted on Hugging Face Datasets at https://huggingface.co/datasets/InstaDeepAI/ms_ninespecies_benchmark.

B. CTMC Discrete Flow Matching Mathematical Foundations

In this section, we provide the concise mathematical foundations for Continuous-Time Markov Chain (CTMC) Discrete Flow Matching on probability simplices, formalizing the framework of Campbell et al. (Campbell et al., 2024).

B.1. Continuous-Time Master Equation and Rate Matrices

Let $\mathcal{X} = \{1, \ldots, S\}$ be a finite categorical state space of size $S = 21$ (20 amino acids + $[M]$). The time evolution of the marginal probability vector $\mathbf{p}_t \in \Delta^{S-1}$ under a continuous-time Markov chain is governed by:

$$\frac{d}{dt}\mathbf{p}_t = \mathbf{p}_t R_t \tag{17}$$

where $R_t \in \mathbb{R}^{S\times S}$ is the transition rate matrix satisfying $R_t(j \mid i) \geq 0$ for $i \neq j$, and diagonal elements $R_t(i \mid i) = -\sum_{j\neq i} R_t(j \mid i)$.

B.2. Conditional Jump-Minimal Dynamics

Given target token $y \in \mathcal{V}_{\text{AA}}$ at $t = 1$ and initial state $[M]$ at $t = 0$, the masked conditional probability path is:

$$p_{t|1}(x \mid y) = \big(1 - \kappa(t)\big)\delta_{[M]}(x) + \kappa(t)\delta_y(x) \tag{18}$$

where $\kappa(t)$ is a smooth schedule with $\kappa(0) = 0, \kappa(1) = 1$. Differentiating with respect to t:

$$\frac{d}{dt}p_{t|1}(x \mid y) = -\dot{\kappa}(t)\delta_{[M]}(x) + \dot{\kappa}(t)\delta_y(x) \tag{19}$$

Under jump-minimal transition dynamics, transitions occur directly from $[M]$ to y:

$$R_t(y \mid [M], y) = \frac{\dot{\kappa}(t)}{1 - \kappa(t)} \tag{20}$$

For any other pair (i, j) with $i \neq [M]$ or $j \neq y$, $R_t(j \mid i, y) = 0$.

B.3. Simulation-Free Marginal Vector Field

Marginalizing over target sequences $\mathbf{y} \sim q(\mathbf{y} \mid \mathcal{S})$, the marginal transition rate from mask $[M]$ to residue $a \in \mathcal{V}_{\text{AA}}$ is:

$$R_t(a \mid [M], \mathcal{S}) = \sum_y R_t(a \mid [M], y) q(y \mid x = [M], \mathcal{S}) \tag{21}$$

$$= \frac{\dot{\kappa}(t)}{1 - \kappa(t)}\hat{p}_{1|t}(a \mid \mathbf{x}_t, \mathcal{S}) \tag{22}$$

where $\hat{p}_{1|t}(a \mid \mathbf{x}_t, \mathcal{S})$ is the neural posterior predicted by the discrete flow decoder. Training by minimizing cross-entropy over masked positions (Equation (10)) directly minimizes the KL divergence between the true velocity field and the parameterized model.

C. Combinatorial KnapsackDP Exactness and Complexity

C.1. Formulation and Exactness

Let $\mathcal{W} = \{\lfloor m(a)/\Delta m \rceil \mid a \in \mathcal{V}_{\text{AA}}\} \subset \mathbb{N}$ denote the set of integer-quantized amino acid masses on grid $\Delta m = 0.02\,\text{Da}$. Let $T[k, b] \in \{0, 1\}$ be the boolean dynamic programming table defined in Equation (11).

Proposition C.1 (Reachability Exactness)**.** *For any positive integer $k \leq K_{\max}$ and mass bin $b \leq B_{\max}$, $T[k, b] = 1$ if and only if there exists a multiset of amino acids $\{a_1, \ldots, a_k\} \subseteq \mathcal{V}_{AA}$ such that $\sum_{i=1}^{k} \lfloor m(a_i)/\Delta m \rceil = b$.*

Proof. By mathematical induction on residue count k. For $k = 0$, $T[0, 0] = 1$ and $T[0, b] = 0$ for all $b > 0$, which holds trivially. Assume the proposition holds for $k - 1$. For step k:

$$T[k, b] = \bigvee_{a\in\mathcal{V}_{\text{AA}}} T[k - 1, b - w_a] \tag{23}$$

Algorithm 2 Precomputed GPU `KnapsackDP` Table Construction

Input: Residue masses $\{m(a)\}_{a\in\mathcal{V}_{\text{AA}}}$, grid resolution $\Delta m = 0.02\,\text{Da}$, max mass $M_{\max} = 5000\,\text{Da}$, max length $K_{\max} = 50$, instrument tolerance $\tau = 0.1\,\text{Da}$.
Initialize bin count $B_{\max} \leftarrow \lfloor M_{\max}/\Delta m \rfloor = 250{,}000$.
Allocate boolean table $T \in \{0,1\}^{(K_{\max}+1)\times B_{\max}} \leftarrow 0$; set $T[0,0] \leftarrow 1$.
for $k = 1$ **to** $K_{\max}$ **do**
  for $a \in \mathcal{V}_{\text{AA}}$ **do**
    $w_a \leftarrow \lfloor m(a)/\Delta m \rceil$
    $T[k, w_a : B_{\max}] \leftarrow T[k, w_a : B_{\max}] \vee T[k-1, 0 : B_{\max} - w_a]$
  end for
end for
Window size $W \leftarrow 2\lfloor \tau/\Delta m \rfloor + 1 = 11$.
Apply GPU 1D max-pooling with kernel size W:
$T_{\text{pooled}}[k,b] \leftarrow \max_{|j-b|\le 5} T[k,j]$.
Output: Pooled reachability tensor T_{pooled} in GPU memory.

$T[k,b] = 1 \iff \exists a \in \mathcal{V}_{\text{AA}}$ such that $T[k-1, b-w_a] = 1$. By the inductive hypothesis, $T[k-1,b-w_a] = 1 \iff \exists\{a_1,\ldots,a_{k-1}\}$ such that $\sum_{i=1}^{k-1} w_{a_i} = b - w_a$. Setting $a_k = a$ yields $\sum_{i=1}^{k} w_{a_i} = b$. Thus, $T[k,b] = 1$ if and only if the mass bin b is reachable by exactly k residues. □

C.2. Computational Complexity

- **Precomputation:** Computing table T requires $K_{\max} \times |\mathcal{V}_{\text{AA}}| \times B_{\max} = 50 \times 20 \times 250{,}000 = 2.5 \times 10^8$ boolean operations, executing in under 0.25 seconds on GPU once at initialization.
- **Inference Lookup:** At each flow step k, checking chemical admissibility (Equation (13)) consists of a single memory indexing operation $T_{\text{pooled}}[u-1, b]$ per residue candidate. This requires $\mathcal{O}(1)$ time per residue, introducing negligible overhead (< 1.8 ms per batch of 512 spectra).

D. Exhaustive Multi-Species Cross-Kingdom Generalization

To evaluate whether the learned continuous-time vector field generalizes across phylogenetic diversity without species-level overfitting, Table 5 reports the complete evaluation of DFlowNovo across all 2,322,730 spectra of the Nine-Species dataset. Pretrained exclusively on human MassIVE-KB spectra, non-human species represent zero-shot cross-organism evaluation.

Species Display Name	Taxonomic Group	Spectrum Count	Peptide Acc. (%)	Mass Match (%)	AA F1 (%)	AA Precision (%)	AA Recall (%)	Length Acc. (%)
Mus musculus (Mouse)	Mammalia	25,963	48.06	51.94	75.02	75.35	74.69	76.86
Homo sapiens (Human)	Mammalia	43,416	65.14	67.86	78.60	78.76	78.44	84.98
Candidatus endoloripes	Bacteria	79,144	48.98	52.57	71.18	71.46	70.91	75.09
Vigna mungo (Black gram)	Plantae	107,955	69.15	73.47	86.96	87.06	86.86	88.08
Saccharomyces cerevisiae (Yeast)	Fungi	104,163	66.44	68.24	81.16	81.28	81.04	84.77
Solanum lycopersicum (Tomato)	Plantae	177,721	69.13	71.94	81.93	82.04	81.82	86.81
Apis mellifera (Honeybee)	Insecta	192,397	55.01	58.75	72.12	72.31	71.93	78.69
Methanosarcina mazei	Archaea	262,739	62.59	66.57	78.40	78.47	78.33	83.04
Bacillus subtilis	Bacteria	1,329,232	65.83	68.78	77.72	77.89	77.56	84.29
Macro-Average / Total	**6 Biological Kingdoms**	**2,322,730**	**61.15**	**64.46**	**78.12**	**78.29**	**77.95**	**82.25**

Table 5. **Comprehensive Per-Species Extended Performance on all Nine Species ($N = 2{,}322{,}730$ spectra total).** Across all nine organisms, DFlowNovo recovers 224,249 unique peptides with a macro-average Amino Acid F1 of 78.12% and Peptide Accuracy of 61.15%.